Bharati Vidyapeeth
(Deemed to be University)

Institute of Management and Research, New Delhi.

Lesson Plan
for

Lab on Web Programming with Project

BCA -III (Sem – VI)
(CBCS 2022)

Course Code: 605	Type: DSC	Credits: 2	Evaluation: UE:IE → 60:40

Prepared By
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Lab on Web Programming (PHP)

1. **Programme Outcomes (POs):**

On completion of BCA (Honors) Four Year Degree Programme the expected programme outcomes that a studentshould be able to demonstrate are the following:

PO1. Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.

PO2. Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.

PO3. Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand and propose integrated solutions using emerging technologies.

PO4. Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well informed conclusions.

PO5. Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions

PO6. Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.

PO7. Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.

PO8. Project Management: Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.

PO9. Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.

PO10. Societal & Environmental Concern: Ability to recognize economical, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.

PO11. Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment.

PO12. Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

2. **Course Objective:**

- To get knowledge of dynamic web site development
- To make students able to design, develop the various types of web based applications.
- To get student familiar with various functionality of PHP

3. **Course Outcome:**

CO1: To apply concept of array, looping, function,, file handling, string function

CO2: To create form with basic functionality

CO3: To create Database and Tables with SQL and create Database connectivity

CO4: To create website with implementation of all concepts

4. CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	-	-	-	-	-	-	-	2	-
CO2	2	2	-	-	-	-	-	-	-	-	-	-
CO3	2	-	-	-	1	-	-	-	-	-	-	-
CO4	2	-	2	-	2	-	-	-	-	-	1	-

5. Assessment Methods

Component	**Description**	**Weightage**
Internal	First internal question paper will be based on first 75% of syllabus	20 marks
CES 1	Class Test	5 marks
CES 2	Practical File, GitHub Repository and Project Assessment	5 marks
Attendance	As per New Attendance policy	10 marks

6. Detailed Lesson Plan

SN	Topic (Exact Syllabus Wording)	CO	Pedagogy	Bloom's Level
1	Write a Program for finding the biggest number in an array without using any array functions.	CO1	Lab Demo + Hands-on	Understand/Apply
2	Write a program to square of a number.	CO1	Lab Demo + Hands-on	Understand/Apply
3	Write a program to print Factorial of any number.	CO1	Lab Demo + Hands-on	Understand/Apply
4	Write a program in PHP to print Fibonacci series.	CO1	Lab Demo + Hands-on	Understand/Apply
5	Write a program to find whether a number is Armstrong or not.	CO1	Lab Demo + Hands-on	Understand/Apply
6	Write a program to find HCF of two numbers.	CO1	Lab Demo + Hands-on	Understand/Apply
7	Write a program to demonstrate four built in functions.	CO1	Lab Demo + Hands-on	Understand/Apply
8	Program to print the given star format.	CO1	Lab Demo + Hands-on	Understand/Apply
9	Write a program to make a chess.	CO1	Lab Demo + Hands-on	Understand/Apply
10	Unit 1 Practical Assessment.	CO1	Lab Demo + Hands-on	Understand/Apply
11	Create the form and print message based on user selection.	CO2	Lab Demo + Hands-on	Understand/Apply
12	Write a PHP program to create and manage a database using SQL commands.	CO2	Lab Demo + Hands-on	Understand/Apply
13	Write a PHP program to create and validate an email id.	CO2	Lab Demo + Hands-on	Understand/Apply

14	Write a PHP script to accept personal details of student (Page 1).	CO2	Lab Demo + Hands-on	Understand/Apply
15	Accept marks of six subjects (Page 2).	CO2	Lab Demo + Hands-on	Understand/Apply
16	Print marklist (Page 3).	CO2	Lab Demo + Hands-on	Understand/Apply
17	Using PHP and SQL, create and validate a sample login form.	CO2	Lab Demo + Hands-on	Understand/Apply
18	Output form with username and password fields.	CO2	Lab Demo + Hands-on	Understand/Apply
19	Validate username-password pair from database.	CO2	Lab Demo + Hands-on	Understand/Apply
20	Unit 2 Practical Assessment.	CO2	Lab Demo + Hands-on	Understand/Apply
21	Make a connection to MySQL database using include/require.	CO3	Lab Demo + Hands-on	Understand/Apply
22	Create database TEST if it does not exist.	CO3	Lab Demo + Hands-on	Understand/Apply
23	Select the TEST database.	CO3	Lab Demo + Hands-on	Understand/Apply
24	Create table USER exerciseusers if it does not exist.	CO3	Lab Demo + Hands-on	Understand/Apply
25	Define USERNAME as UNIQUE field.	CO3	Lab Demo + Hands-on	Understand/Apply
26	Error handling if any DB operation fails.	CO3	Lab Demo + Hands-on	Understand/Apply
27	Design web page that accepts username and password.	CO3	Lab Demo + Hands-on	Understand/Apply
28	Authenticate username and password using PHP.	CO3	Lab Demo + Hands-on	Understand/Apply
29	Session management implementation.	CO3	Lab Demo + Hands-on	Understand/Apply
30	Logout functionality.	CO3	Lab Demo + Hands-on	Understand/Apply

31	File handling using PHP – Write operation.	CO4	Lab Demo + Hands-on	Understand/Apply
32	File handling using PHP – Read operation.	CO4	Lab Demo + Hands-on	Understand/Apply
33	Display file contents.	CO4	Lab Demo + Hands-on	Understand/Apply
34	Integrated Form + Database practical.	CO4	Lab Demo + Hands-on	Understand/Apply
35	Security validation improvements.	CO4	Lab Demo + Hands-on	Understand/Apply
36	Debugging database connectivity.	CO4	Lab Demo + Hands-on	Understand/Apply
37	Mini Project Discussion.	CO4	Lab Demo + Hands-on	Understand/Apply
38	Guided Practical – Website integration.	CO4	Lab Demo + Hands-on	Understand/Apply
39	Revision – Unit 1 & 2.	CO4	Lab Demo + Hands-on	Understand/Apply
40	Revision – Unit 3.	CO4	Lab Demo + Hands-on	Understand/Apply
41	Revision – Unit 4.	CO4	Lab Demo + Hands-on	Understand/Apply
42	Full Practical Mock Test.	CO4	Lab Demo + Hands-on	Understand/Apply
43	Project Implementation.	CO4	Lab Demo + Hands-on	Understand/Apply
44	Project Testing & Validation.	CO4	Lab Demo + Hands-on	Understand/Apply
45	Final Practical + Viva.	CO4	Lab Demo + Hands-on	Understand/Apply

7. Textbooks:

Sr. No.	Name of the Author	Title of the Book	Edition & Year	Publisher Company
1	Welling Thomson	PHP and MySQL Web Development	Fifth Edition, 2017	Pearson Publication

8. Reference Books:

Sr. No.	Name of the Author	Title of the Book	Edition & Year	Publisher Company
1	Robin Nixon	Learning PHP, MySQL & JavaScript	7th Edition, 2025	Oreilly Publication

9. Online Resources:

- https://www.geeksforgeeks.org/php/php-tutorial/
- https://leetcode.com/problemset/
- https://www.w3schools.com/MySQL/default.asp

10. MOOC Courses:

- PHP and MySQL By Prof Kannan Moudgalya - Principal Investigator of Spoken Tutorial Project, Indian Institute of Technology Bombay https://onlinecourses.swayam2.ac.in/aic20_sp32/preview

11. Research Articles:

- K. Bounnady, K. Phanthavong, S. Pathoumvanh and K. Sihalath, "Comparison the processing speed between PHP and ASP.NET," 2016 13th International Conference on Electrical Engineering/ Electronics, Computer, Telecommunications and Information Technology (ECTI-CON), Chiang Mai, Thailand, 2016, pp. 1-5, doi: 10.1109/ECTICon.2016.7561484.
- S. Vimal, R. Harshini, D. Sankari and D. Srinithi, "AI-Powered Bilingual Chatbot for Promoting Mental Health Awareness in Rural Communities," *2025 10th International Conference on Smart Structures and Systems (ICSSS)*, Chennai, India, 2025, pp. 1-10, doi: 10.1109/ ICSSS66939.2025.11346329.

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2. Course Objective:

- To get knowledge of dynamic web site development
- To make students able to design, develop the various types of web-based applications.
- To get student familiar with various functionality of PHP

3. Course Outcome:

CO1: To study the basics of PHP language as control structures, array, function, strings andfile

handling.
CO2: To understand the concept of cookie and session.
CO3: To under the MYSQL components, and Database connectivity.
CO4: To create website with implementation of all concepts.

4. Syllabus:

Unit	Content	Sessions (Hrs)	COs Number	Teaching Methodology	Cognition Level	Evaluation Tools
1	**Introduction To PHP:** Installing and configuring PHP, Building blocks of PHP:PHP tags, variables, datatypes, operators, expressions, constants, Control Structures:conditional statements, loops,	9	CO 1	Lecture with Demonstrations	Understand	Quiz Short Answers
2	**Working With Functions And Arrays:** Working with functions: What is a function? Function declaration and definition, Calling function, user defined functions, variable	9	CO 1	Lecture with Demonstrations	Understand	Quiz, Short Answers

3	**String Manipulation:** Working with strings, Dates and Time: Formatting, investigating and manipulating strings with PHP, using date and time functions in PHP, Working with forms: Creating a simple input form. File Handling: Saving data, storing and retrieving Bob's order, processing files, opening file, writing to a file,	9	CO 1	Lecture with Demonstrations	Understand, Apply	Quiz Short Answers
4	**Working With Cookies and Sessions :** Working with cookies: Introducing cookies, setting and deleting cookies with PHP. Working with session: starting a session, working with session variables, passing session IDs in the query string, destroying sessions and unsetting	9	CO2	Lecture with Demonstrations	Understand, Apply	Quiz Short Answers

5	**MYSQL :** Creating web database: Using MySQL monitor, logging into MySQL, creating databases and users, setting users and privileges, column data types **Working with MySQL database**: Inserting data into database, retrieving data from the database, retrieving data with specific criteria, retrieving data from multiple tables, retrieving data in particular order, grouping and aggregate data, using sub queries	9	CO3, CO4	Lecture with Demonstrations	Create, Apply	Quiz Short Answers

5. CO-PO Mapping

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO8	PO9	PO 10	PO 11	PO 12
CO 1	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO 2	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO 3	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
CO 4		**2**	**2**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

1. Low , 2- Medium, 3- High, "-" no correlation.

6. Assessment Methods

Component	Description	Weightage
Internal	First internal question paper will be based on first 75% of syllabus	20 marks
CES 1	Class Test	5 marks
CES 2	Based on Mini Projects Combined file 1 to 5	5 marks
Attendance	As per New Attendance policy	10 marks

7. Detailed Lesson Plan

SN.	Topic	Subtopic	COs	Pedagogy	Book Reference/Text	Hrs.	Mins.	ONo.
1	Introduction to PHP	Introduction to PHP & Web Concepts	CO1	Lecture with Demonstrations (PPT No. 1-3)	PHP and MySQL Web Development (Ed. -V) by Welling	0	55	1
2		Installing & Configuring PHP	CO1	Lecture with Demonstrations (PPT No. 4-7)	PHP and MySQL Web Development by Welling Thomson,	0	55	2
3		PHP Syntax, Tags & Variables	CO1	Live Coding Demonstrations (PPT No. 8-10)	PHP and MySQL Web Development by Welling Thomson,	0	55	3
4		Data Types & Constants	CO1	Lecture with Demonstrations (PPT No. 11-14)	PHP and MySQL Web Development by Welling Thomson,	0	55	4
5		Operators & Expressions	CO1	Lecture with Demonstrations (PPT No. 15-21)	PHP and MySQL Web Development by Welling Thomson,	0	55	5
6		Conditional Statements	CO1	Lecture with Demonstrations (PPT No. 22-24)	PHP and MySQL Web Development by Welling Thomson,	0	55	6
7		Looping Statements	CO1	Lecture with Demonstrations (PPT No. 25-27)	PHP and MySQL Web Development by Welling Thomson,	0	55	7

8		Switch Statement	CO1	Lecture with Demonstrations (PPT No. 28)	PHP and MySQL Web Development by Welling Thomson,	0	55	8
9		Case Study -1	CO1	Scenario Based Problem Solving, Mini Project–1: PHP Order Processing	https://leetcode.com/problems/roman-to-integer/	0	55	9
10	Functions & Arrays	Introduction to Functions	CO1	Lecture with Demonstrations (PPT No. 29-30)	PHP and MySQL Web Development by Welling Thomson,	0	55	10
11		Function Calling, Parameteres	CO1	Lecture with Demonstrations (PPT No. 31-33)	PHP and MySQL Web Development by Welling Thomson,	0	55	11
12		User Defined Functions	CO1	Lecture with Demonstrations (PPT No. 33-34)	PHP and MySQL Web Development by Welling Thomson,	0	55	12
13		Variable Scope	CO1	Lecture with Demonstrations (PPT No. 35)	PHP and MySQL Web Development by Welling Thomson,	0	55	13
14		Introduction to Arrays	CO1	Lecture with Demonstrations (PPT No. 36-37)	PHP and MySQL Web Development by Welling Thomson,	0	55	14
15		Indexed & Associative Arrays	CO1	Lecture with Demonstrations (PPT No. 38-39)	PHP and MySQL Web Development by Welling Thomson,	0	55	15
16		Sorting & Reordering Arrays	CO1	Lecture with Demonstrations (PPT No. 40-43)	PHP and MySQL Web Development by Welling Thomson,	0	55	16
17		PHP Classes – Introduction	CO1	Lecture with Demonstrations (PPT No. 44-48)	PHP and MySQL Web Development by Welling Thomson,	0	55	17

18		Case Study -2 Mini Project–2:	CO1	Scenario Based Problem Solving. Mini Project–2: Reusable PHP Utility Library using	https://leetcode.com/problems/median-of-two-sorted-arrays/	0	55	18
19	Strings & File Handling	Strings in PHP	CO1	Lecture with Demonstrations (PPT No. 49-50)	PHP and MySQL Web Development by Welling Thomson,	0	55	19
20		String Manipulation Functions	CO1	Lecture with Demonstrations (PPT No. 51-53)	PHP and MySQL Web Development by Welling Thomson,	0	55	20
21		Date & Time Functions	CO1	Lecture with Demonstrations (PPT No. 54-56)	PHP and MySQL Web Development by Welling Thomson,	0	55	21
22		Working with Forms	CO1	Lecture with Demonstrations (PPT No. 57-58)	PHP and MySQL Web Development by Welling Thomson,	0	55	22
23		Form Validation	CO1	Lecture with Demonstrations (PPT No. 59-62)	PHP and MySQL Web Development by Welling Thomson,	0	55	23
24		Introduction to File Handling	CO1	Lecture with Demonstrations (PPT No. 63-64)	PHP and MySQL Web Development by Welling Thomson,	0	55	24
25		Reading from Files	CO1	Lecture with Demonstrations (PPT No. 65)	PHP and MySQL Web Development by Welling Thomson,	0	55	25
26		Writing to Files	CO1	Lecture with Demonstrations (PPT No. 66)	PHP and MySQL Web Development by Welling Thomson,	0	55	26
27		Case Study- 3 Mini Project–3:		Scenario Based Problem Solving. Mini Project–3: Form Submission	https://www.geeksforgeeks.org/php/php-form-processing/	0	55	27

28	Cookies & Sessions	Introduction to Cookies	CO2	Lecture with Demonstrations (PPT No. 67)	PHP and MySQL Web Development by Welling Thomson,	0	55	28
29		Creating & Accessing Cookies	CO2	Lecture with Demonstrations (PPT No. 68-69)	PHP and MySQL Web Development by Welling Thomson,	0	55	29
30		Deleting Cookies	CO2	Lecture with Demonstrations (PPT No. 70)	PHP and MySQL Web Development by Welling Thomson,	0	55	30
31		Introduction to Sessions	CO2	Lecture with Demonstrations (PPT No. 71)	PHP and MySQL Web Development by Welling Thomson,	0	55	31
32		Session Variables	CO2	Lecture with Demonstrations (PPT No. 72)	PHP and MySQL Web Development by Welling Thomson,	0	55	32
33		Passing Session IDs	CO2	Lecture with Demonstrations (PPT No. 73)	PHP and MySQL Web Development by Welling Thomson,	0	55	33
34		Destroying Sessions	CO2	Lecture with Demonstrations (PPT No. 74)	PHP and MySQL Web Development by Welling Thomson,	0	55	34
35		Applications of Sessions	CO2	Lecture with Demonstrations (PPT No. 75)	PHP and MySQL Web Development by Welling Thomson,	0	55	35
36		Case Study – 4 Mini Project - 4		Scenario Based Problem Solving Mini Project – 4 Demonstrating Cookies Session for	https://www.geeksforgeeks.org/php/php-sessions/	0	55	36
37		Introduction to MySQL	CO3, CO4	Lecture with Demonstrations (PPT No. 76-78)	PHP and MySQL Web Development by Welling Thomson,	0	55	37
38		Creating Databases & Users	CO3, CO4	Lecture with Demonstrations (PPT No. 79-80)	PHP and MySQL Web Development by Welling Thomson,	0	55	38

39	MySQL with PHP	Table Creation & Data Types	CO3, CO4	Lecture with Demonstrations (PPT No. 81-82)	PHP and MySQL Web Development by Welling Thomson,	0	55	39
40		Inserting and Retrieving Data	CO3, CO4	Lecture with Demonstrations (PPT No. 83-84)	PHP and MySQL Web Development by Welling Thomson,	0	55	40
41		Conditional Retrival	CO3, CO4	Lecture with Demonstrations (PPT No. 85-86)	PHP and MySQL Web Development by Welling Thomson,	0	55	41
42		Subqueries & Aggregates	CO3, CO4	Lecture with Demonstrations (PPT No. 87-90)	PHP and MySQL Web Development by Welling Thomson,	0	55	42
43		Updating & Deleting Records	CO3, CO4	Lecture with Demonstrations (PPT No. 91-93)	PHP and MySQL Web Development by Welling Thomson,	0	55	43
44		Dropping Tables & Databases	CO3, CO4	Lecture with Demonstrations (PPT No. 93-95)	PHP and MySQL Web Development by Welling Thomson,	0	55	44
45		Case Study -5 Mini Project: 5		Scenario Based Problem Solving Mini Project: 5 Student Record Management	https://www.squareboat.com/clients/holidays-wallet	0	55	45

 # Scenario based problem solving from online platforms like Leetcode, Geekes for Geeks and Squareboat improves students to apply their learning towards real words cases.

8. Mini Projects:

I. Order Processing System :

Develop an HTML form to collect customer and product information.
Implement logic to calculate total amount (price × quantity).
Create an admin page to view all stored orders.
Apply basic validation and security measures (e.g., required fields).
Test the complete workflow from order placement to data storage.

II.Reusable PHP Utility Library :

Develop reusable PHP functions.
Implement arrays for configuration and data handling.
Design utility classes using OOP concepts.

Create a modular structure for easy integration into other projects.
Improve clean coding and structured programming practices.

III.Form Submission and File Logging System:

Develop a user input form using HTML.
Process form data using PHP.
Validate and sanitize user inputs.
Store submitted data into a structured log file (.txt or .log).
Implement file handling operations (open, write, append, close).
Display confirmation message after successful submission.

IV.Demonstrating Cookies & Sessions for Order Processing System:

Design order form.
Implement session-based cart storage.
Set and retrieve cookies for user preference.
Display dynamic order summary.
Clear session after order confirmation.
Test cookie persistence across browser refresh.

V.Student Record Management System:

Design database and student table.
Implement add, view, edit, delete functionality.
Create dynamic interface using PHP loops.
Validate user input.
Test system for multiple entries.

9. Textbooks:

Sr. No.	Name of the Author	Title of the Book	Edition & Year	Publisher Company
1	Welling Thomson	PHP and MySQL Web Development	Fifth Edition,	Pearson Publication

10.Reference Books:

Sr. No.	Name of the Author	Title of the Book	Edition & Year	Publisher Company
1	Robin Nixon	Learning PHP, MySQL & JavaScript	7th Edition, 2025	Oreilly Publication

11. Online Resources:

- https://www.geeksforgeeks.org/php/php-tutorial/
- https://leetcode.com/problemset/
- https://www.w3schools.com/MySQL/default.asp

12. MOOC Courses:

- PHP and MySQL By Prof Kannan Moudgalya - Principal Investigator of Spoken Tutorial Project, Indian Institute of Technology Bombay https://onlinecourses.swayam2.ac.in/aic20_sp32/preview

13. Research Articles:

- K. Bounnady, K. Phanthavong, S. Pathoumvanh and K. Sihalath, "Comparison the processing speed between PHP and ASP.NET," 2016 13th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON), Chiang Mai, Thailand, 2016, pp. 1-5, doi: 10.1109/ECTICon.2016.7561484.
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